

Bill Verification and Approval System (BVAS)

For Monthly Distribution Bills Processing

1. Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional, non-functional, security, and integration requirements for developing a secure, auditable, and role-based web application named **Bill Verification and Approval System (BVAS)**.

The system aims to:

- Vendors to submit monthly distribution bills
- District authorities to verify, approve, or reject bills with digital signatures
- HQ officials to monitor status, generate reports, and manage users

2. Scope

The system covers:

- **Vendor registration & bill submission**

The process begins when a vendor accesses the BVAS web portal and completes self-registration using basic details, with potential headquarters approval required in later phases. Once registered and logged in securely with username/password (and optional 2FA), the vendor reaches the dedicated dashboard. Here, they select the relevant bill month and year and upload the main bill along with supporting documents (in PDF or XLSX format, up to a total of 15 MB), and enter or edit distribution quantities district-wise in a structured table covering all 13 districts of Uttarakhand. After reviewing a final preview of the entire submission, the vendor confirms it, upon which the bill status updates to "Pending – District Verification". Vendors can later track the real-time status, view submission history, and re-submit corrections only if the bill was previously rejected by the district authority.

- **District-level verification with digital signing**

District verifiers, who are authorized government officials at the district level, log in to access a personalized pending queue that displays only those bills assigned to their specific district. For each pending bill, they review detailed information and perform a side-by-side comparison between the quantities submitted by the vendor and the official records from the ePOS system, with any discrepancies automatically highlighted for easy identification. If satisfied after scrutiny, the verifier selects the "Approve" action, which triggers a mandatory digital signing session using either a Class 3 Digital Signature Certificate (via USB token or .pfx/.p12 file) or Aadhaar-based eSign through licensed providers. Before signing, the system requires re-authentication (password or OTP), validates the certificate fully and applies an invisible PAdES signature for tamper-proofing, along with optional visible signature details on the PDF. Upon successful signing, the bill becomes locked as read-only, full signing metadata

(timestamp, IP, certificate serial, signer name) is logged immutably, and the status updates to approved. If issues are found, the verifier can reject the bill but must provide mandatory remarks, allowing the vendor to correct and re-submit.

- **HQ oversight, reporting, and admin functions**

Headquarters administrators, typically senior officers, gain comprehensive, real-time visibility across the entire state through a powerful analytics dashboard featuring KPI cards that display counts of pending, approved, and rejected bills for the current and previous months. They can apply advanced filters (by month range, multiple districts, specific vendors, or status) to drill down into details and generate downloadable reports in Excel format or digitally signed PDF for official records. Administrative functions include full user management — creating, editing, blocking users, and assigning districts/roles — as well as viewing the complete audit trail and action history for every individual bill to ensure transparency and accountability. Additionally, HQ can lock or unlock bill submission windows for specific months to enforce deadlines or handle special situations, enabling centralized monitoring, quick issue resolution, and overall governance of the BVAS platform across all districts and vendors.

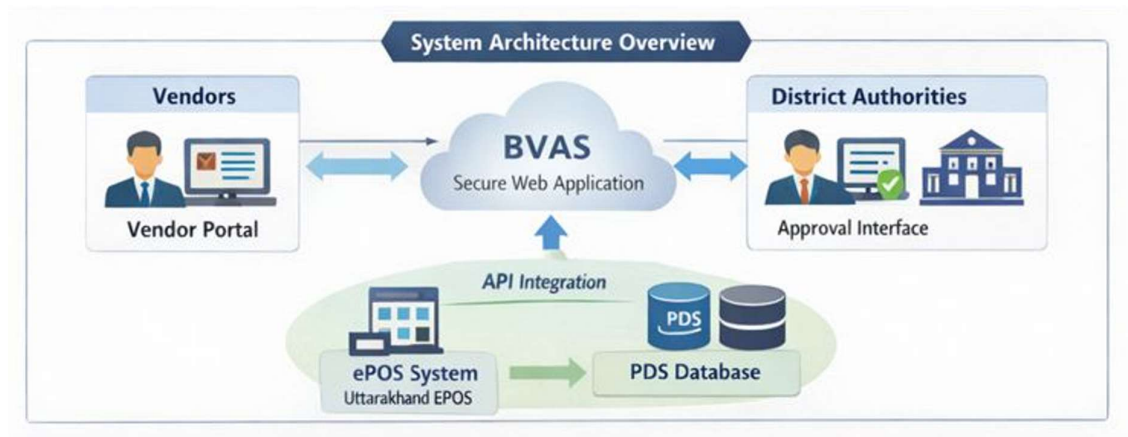
- **Integration with external PDS/ePOS system (epos.uk.gov.in)**

Integration with the Uttarakhand ePOS (Electronic Point of Sale) system forms a critical bridge for data accuracy and efficiency in the bill approval workflow. In the preferred Phase 2 approach, the BVAS establishes secure REST API connectivity using OAuth2/JWT authentication to automatically fetch real-time distribution quantities recorded at fair price shops under various PDS schemes. This data is then displayed directly to vendors during bill preparation (for reference or auto-filling) and to district verifiers during review (for automatic mismatch highlighting), significantly reducing manual data entry errors and verification time. As a Phase 1 fallback until full API readiness, vendors manually upload relevant ePOS reports (in PDF or Excel format) as supporting documents during submission. This phased integration ensures submitted bill quantities align with official transaction records from the ground level, enhancing trust, minimizing disputes, and supporting faster, more reliable approvals in the Public Distribution System ecosystem.

- **Compliance with Indian digital signature standards**

Compliance with Indian digital signature standards is embedded throughout the approval workflow to guarantee legal validity, integrity, and non-repudiation of every approved bill. All final approvals by district verifiers are mandatorily executed using either Class 3 Digital Signature Certificates (DSC) — loaded via USB token or .pfx/.p12 files — or Aadhaar-based eSign facilitated through licensed Empanelled Service Providers (such as eMudhra, CDSL, or Capricorn), in full adherence to the **Information Technology Act 2000** and **Controller of Certifying Authorities (CCA)** guidelines. Before any signing session, the system enforces re-authentication and rigorously validates the certificate (checking chain of trust to Root CA India, revocation

status via CRL/OCSP, and expiry). It applies an invisible PAdES-level signature (preferably B-LT for long-term validation and tamper detection), stores the resulting immutable signed PDF along with document hash and signature metadata, and logs all relevant details. This robust mechanism ensures that digitally approved bills hold the same legal weight as physically signed documents, enabling a completely paperless, secure, and auditable process compliant with Government of India e-governance standards.



3. Overview

BVAS is a web-based, role-driven system accessible through standard browsers over HTTPS.

- **Security:** Digital Signatures (DSC, eSign), Role-Based Access Control (RBAC)
- **Auditability:** Immutable Logs, Signed PDFs
- **Transparency:** Real-Time Status Tracking
- **Compliance:** Government of India Standards

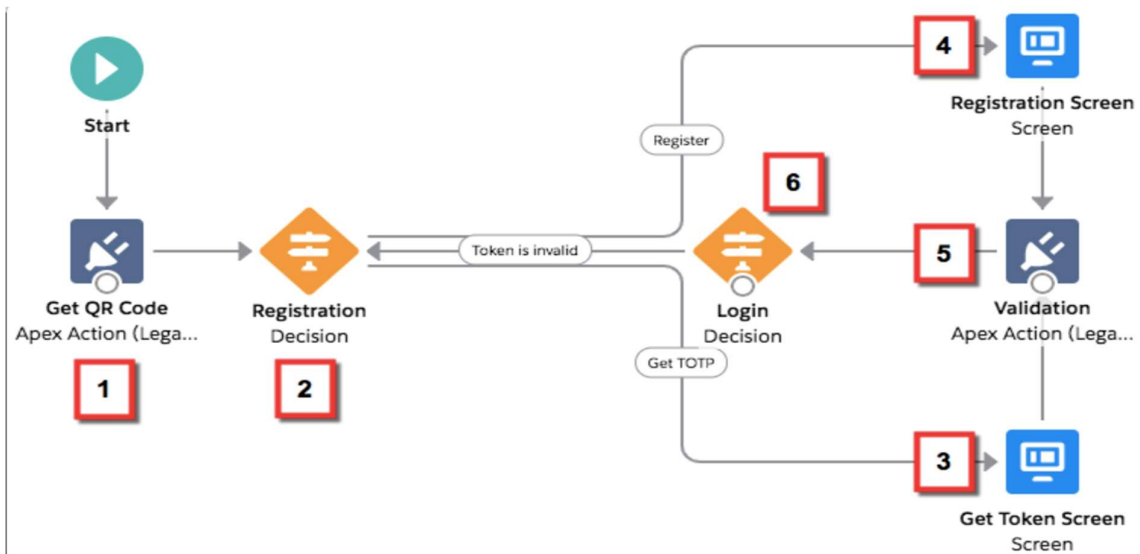


4. Specific Requirements – Functional

4.1 Authentication Module

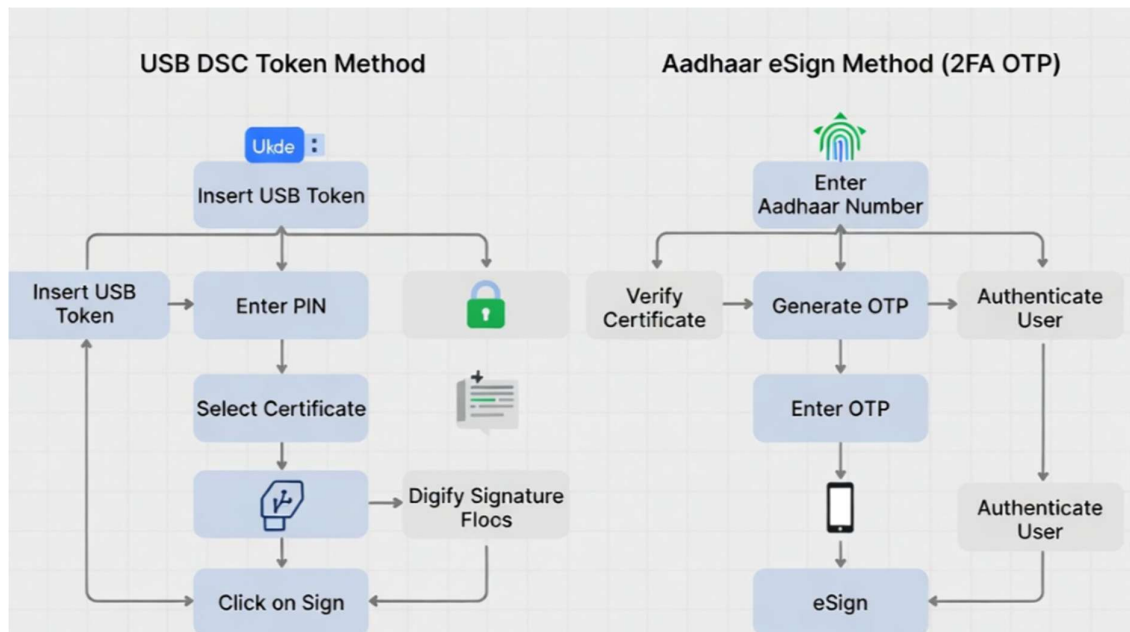
- Username + Password + optional 2FA (TOTP – Google Authenticator, Authy, etc.)

- **Password Policy:** Min. 12 characters, 1 uppercase, 1 lowercase, 1 digit, 1 special character
- CAPTCHA after 3 failed attempts
- Password reset via OTP to registered mobile (Aadhaar linked preferred)
- Role-Based Access Control (RBAC) – dynamic permissions
- Optional vendor self-registration with HQ approval workflow



4.2 Digital Signature & eSign Module

- Class 3 DSC (USB token / .pfx/.p12 file)
- Aadhaar-linked eSign (2FA OTP flow – IT Act 2000 & CCA Guidelines)
- Validation: CRL, OCSP, chain to Root CA
- Visible signature on PDF (name, designation, date, authority)
- Invisible/embedded signature for tamper detection (mandatory on final documents)
- Immutable storage of document hash + signature



4.3 Integration with ePOS (epos.uk.gov.in)

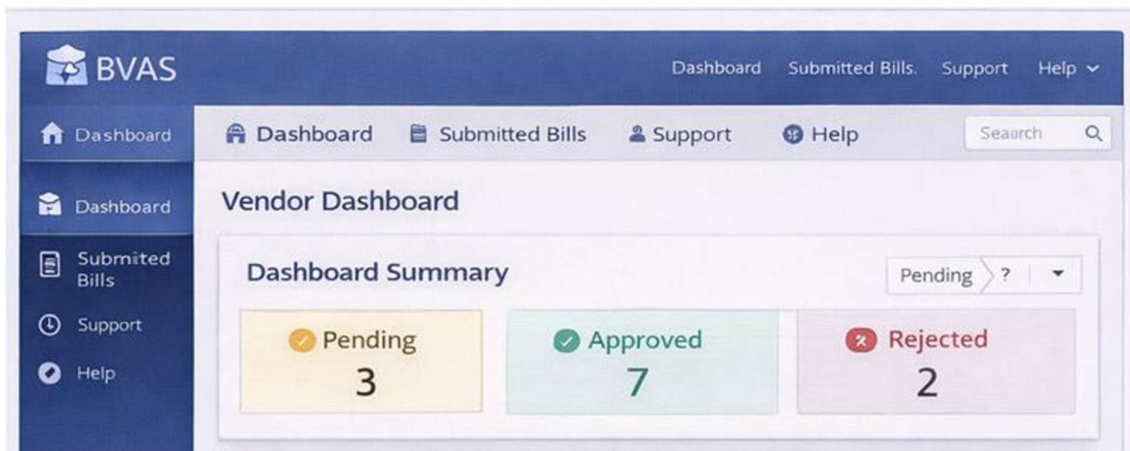
- REST API
- GET /transactions/monthly-summary?district=13&month=2025-12&fps_code=XXXX
- GET /commodity-dispatch?month=YYYY-MM&district=XX

4.4 Main Screens

4.4.1 Screen 1 – Vendor Dashboard

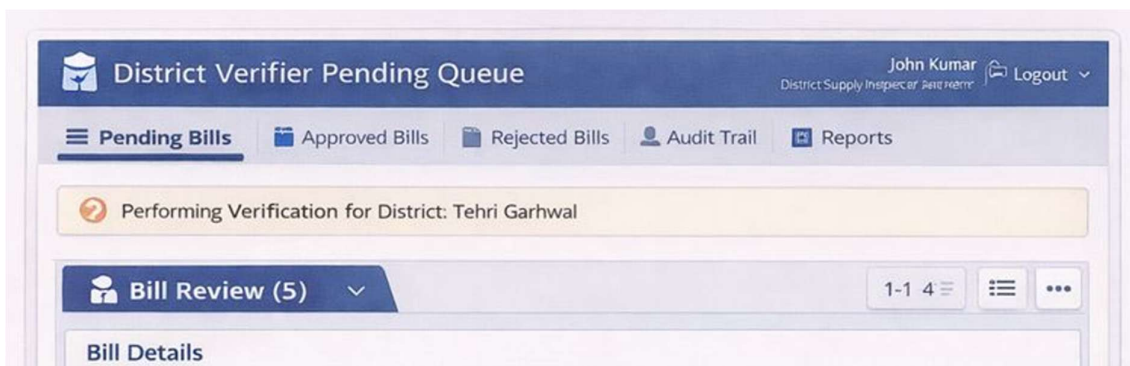
Central hub for vendors after login

- Submit new bill (PDF/XLSX)
- Track status (Pending → Approved → Rejected)
- Re-submit corrected bills after rejection



4.4.2 Screen 2 – District Verifier Pending Queue

- Pending bills queue (district-specific)
- Side-by-side comparison: Vendor data vs ePOS data
- Approve (with digital signature) / Reject (with remarks)
- Complete audit trail logging



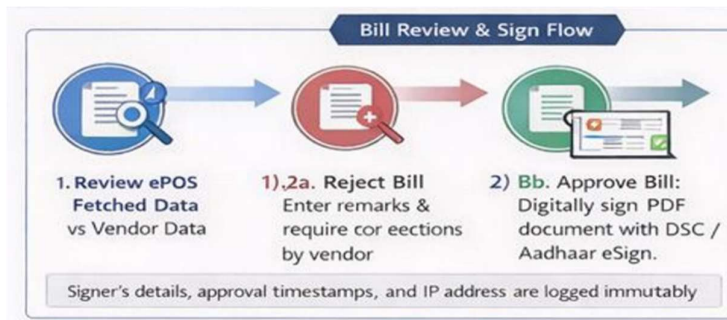
FPS Vs. Vendor data					Vendor Submitted Data	
Commodity	FPS	Data	Settled		View FPS Data & Vendor Data	Status
Wheat	150	150	250	70	View FPS Data	Verify >
Rice	250	250	240	70	155 240	View FPS Data Verify >
Sugar	70	150	235	80	150 235	View FPS Data Reject >

4.4.3 Screen 3 – Bill Review & Sign

Detailed bill review interface with:

- ePOS vs Vendor data comparison
- Approve → Digital Sign (DSC/eSign)

- *Reject → Remarks mandatory*



BVAS

Bill Integration

Dist Name:	Buldhara	Transfer Date:
Remitter:	Condel	Transfer Date:
Hub Code:	100000	Transfer Date:

Rejection

Enter rejection reason...

Approve - Digitally Sign **Reject**

2 *John Kumar*

Name: John Kumar
Designation: District Supply Inspector
Serial Number: 2011/2024
Authority entered: e-Mudhra

4.4.4 Screen 4 – HQ Analytics Dashboard

- Overall + district-wise status overview
- Filters: district, vendor, status
- Downloadable reports (PDF/Excel)
- Admin functions: lock/unlock months, user & role management
- Immutable audit trails

HQ Analytics Dashboard Download Report

Overview

8 Pending 21 Approved 3 Rejected

District	Pending	Approved	Rejected
Almora	6	2	2
Chamoli	5	3	4
Champavat	4	1	1
Dehradun	5	1	8
Haridwar	8	1	1
Pauri Garhwal	2	1	1
Tirath Garhwal	8	1	2
Total	6	21	3

Filters:

Dist: 2020 +

Crash book 4

Pain Bathinal 6

Tent Bathinal -2